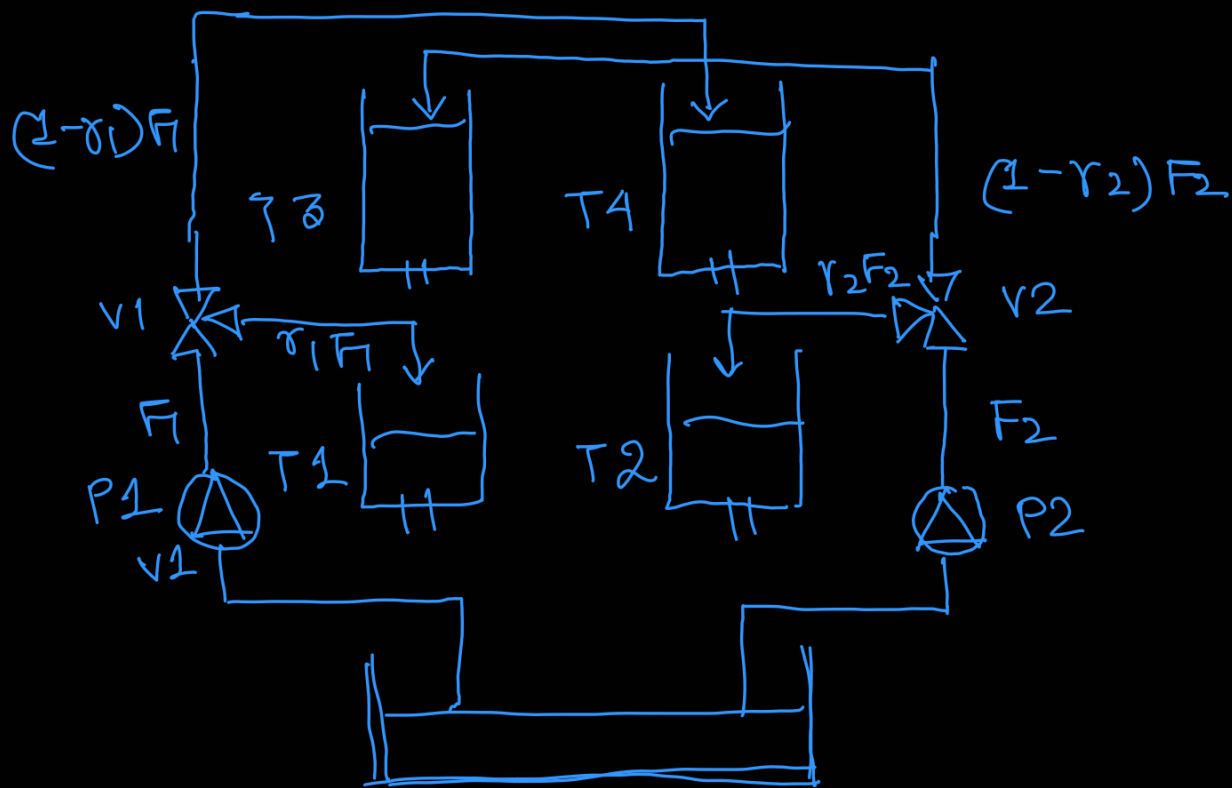


17 Quadruple tank system.



- Derive dynamic model
- Derive non-linear state space model in vector matrix form
- Derive linear state " " "
- Compute state transition matrix using following data

$$A_1, A_3 = 28 \text{ cm}^2 \quad A_2, A_4 = 32 \text{ cm}^2$$

$$a_1, a_3 = 0.071 \text{ cm}^2 \quad a_2, a_4 = 0.057 \text{ cm}^2$$

$$k_1, k_2 = 3.33, 3.35 \quad v_1, v_2 = 3, 3 \quad \gamma_1, \gamma_2 = 0.7, 0.6$$

Soln :

(a) Tank 1,

$$\frac{dV_1}{dt} = A_1 \frac{dh_1}{dt} = r_1 F_1 - a_1 \sqrt{2gh_1} + a_3 \sqrt{2gh_3}$$

$$\frac{dh_1}{dt} = \gamma_1 \frac{F_1}{A_1} - \frac{a_1}{A_1} \sqrt{2gh_1} + \frac{a_3}{A_1} \sqrt{2gh_3} \quad \text{--- (i)}$$

Tank 2,

$$\frac{dh_2}{dt} = \frac{\gamma_2 F_2}{A_2} - \frac{a_2}{A_2} \sqrt{2gh_2} + \frac{a_4}{A_2} \sqrt{2gh_4} \quad \text{--- (ii)}$$

Tank 3,

$$\frac{dh_3}{dt} = \frac{(1-\gamma_2)F_2}{A_3} - \frac{a_3}{A_3} \sqrt{2gh_3} \quad \text{--- (iii)}$$

Tank 4,

$$\frac{dh_4}{dt} = \frac{(1-\gamma_1)F_1}{A_4} - \frac{a_4}{A_4} \sqrt{2gh_4} \quad \text{--- (iv)}$$

where $F_1 = K_1 V_1$, $F_2 = K_2 V_2$,

(b) In vector matrix form, let $u_1 = V_1$, $u_2 = V_2$ be the controlled variables .

$$\frac{dh_1}{dt} = -\frac{a_1}{A_1} \sqrt{2g} \cdot \sqrt{h_1} + \frac{a_3}{A_1} \sqrt{2g} \sqrt{h_3} + \frac{\gamma_1 K_1}{A_1} u_1$$

$$\frac{dh_2}{dt} = -\frac{a_2}{A_2} \sqrt{2g} \cdot \sqrt{h_2} + \frac{a_4}{A_2} \sqrt{2g} \cdot \sqrt{h_4} + \frac{\gamma_2 K_2}{A_2} \cdot u_2$$

$$\frac{dh_3}{dt} = -\frac{a_3}{A_3} \sqrt{2g} \cdot \sqrt{h_3} + \frac{(1-\gamma_2) K_2}{A_3} u_2$$

$$\frac{dh_4}{dt} = -\frac{a_4}{A_4} \sqrt{2g} \cdot \sqrt{h_4} + \frac{(1-\gamma_1) K_1}{A_4} \cdot u_1$$

$$\begin{bmatrix} \dot{h}_1 \\ \dot{h}_2 \\ \dot{h}_3 \\ \dot{h}_4 \end{bmatrix} = \begin{bmatrix} -\frac{a_1}{A_1} \sqrt{2g} & 0 & \frac{a_3}{A_1} \sqrt{2g} & 0 \\ 0 & -\frac{a_2}{A_2} \sqrt{2g} & 0 & \frac{a_4}{A_2} \sqrt{2g} \\ 0 & 0 & -\frac{a_3}{A_3} \sqrt{2g} & 0 \\ 0 & 0 & 0 & -\frac{a_4}{A_4} \sqrt{2g} \end{bmatrix} \begin{bmatrix} \sqrt{h_1} \\ \sqrt{h_2} \\ \sqrt{h_3} \\ \sqrt{h_4} \end{bmatrix} + \begin{bmatrix} \frac{\gamma_1 K_1}{A_1} & 0 \\ 0 & \frac{\gamma_2 K_2}{A_2} \\ 0 & \frac{(1-\gamma_2) K_2}{A_3} \\ \frac{(1-\gamma_1) K_1}{A_4} & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

(c) Linearizing around steady state

$$\sqrt{h_1} = \sqrt{h_1^s} + (h_1 - h_1^s) \cdot \frac{1}{2\sqrt{h_1^s}}$$

$$\sqrt{h_2} = \sqrt{h_2^s} + (h_2 - h_2^s) \cdot \frac{1}{2 \cdot \sqrt{h_2^s}}$$

$$\sqrt{h_3} = \sqrt{h_3^s} + (h_3 - h_3^s) \cdot \frac{1}{2 \cdot \sqrt{h_3^s}}$$

$$\sqrt{h_4} = \sqrt{h_4^s} + (h_4 - h_4^s) \cdot \frac{1}{2 \cdot \sqrt{h_4^s}}$$

$$x_i = h_i - h_i^s \Rightarrow \frac{dx_i}{dt} = \frac{dh_i}{dt} ; y_i = u_i - u_i^s$$

$$\sqrt{h_i} = \sqrt{h_i^s} + \frac{x_i}{2\sqrt{h_i^s}}$$

$$\frac{dx_1}{dt} = -\frac{a_1}{A_1} \sqrt{2g} \left(\sqrt{h_1^s} + \frac{x_1}{2\sqrt{h_1^s}} \right) + \frac{a_3}{A_1} \sqrt{2g} \left(\sqrt{h_3^s} + \frac{x_3}{2\sqrt{h_3^s}} \right) + \frac{\gamma_1 k_1}{A_1} u_1 \quad \text{--- (v)}$$

$$0 = -\frac{a_1 \sqrt{2g}}{A_1} \sqrt{h_1^s} + \frac{a_3 \sqrt{2g}}{A_1} \sqrt{h_3^s} + \frac{\gamma_1 k_1}{A_1} u_1^s \quad \text{--- (vi)}$$

Subtracting (vi) from (v),

$$\frac{dx_1}{dt} = -\frac{a_1}{A_1} \sqrt{\frac{g}{2h_1^s}} x_1 + \frac{a_3}{A_1} \sqrt{\frac{g}{2h_3^s}} x_3 + \frac{\gamma_1 k_1}{A_1} y_1$$

$$\frac{dx_2}{dt} = -\frac{a_2}{A_2} \sqrt{\frac{g}{2h_2^s}} x_2 + \frac{a_4}{A_2} \sqrt{\frac{g}{2h_4^s}} x_4 + \frac{\gamma_2 k_2}{A_2} y_2$$

$$\frac{dx_3}{dt} = -\frac{a_3}{A_3} \sqrt{\frac{g}{2h_3^s}} x_3 + \frac{(1-\gamma_2)k_2}{A_3} y_2$$

$$\frac{dx_4}{dt} = -\frac{a_4}{A_4} \sqrt{\frac{g}{2h_4^s}} x_4 + \frac{(1-\gamma_1)k_1}{A_4} y_1$$

In vector matrix form,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} -\frac{a_1}{A_1} \sqrt{\frac{g}{2h_1^s}} & 0 & \frac{a_3}{A_1} \sqrt{\frac{g}{2h_3^s}} & 0 \\ 0 & -\frac{a_2}{A_2} \sqrt{\frac{g}{2h_2^s}} & 0 & \frac{a_4}{A_2} \sqrt{\frac{g}{2h_4^s}} \\ 0 & 0 & -\frac{a_3}{A_3} \sqrt{\frac{g}{2h_3^s}} & \frac{(1-\gamma_2)k_2}{A_3} \\ 0 & \frac{(1-\gamma_1)k_1}{A_4} & 0 & -\frac{a_4}{A_4} \sqrt{\frac{g}{2h_4^s}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} \frac{\gamma_1 k_1}{A_1} y_1 \\ \frac{\gamma_2 k_2}{A_2} y_2 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} A_1 & 0 & 0 \\ 0 & -\frac{a_2}{A_2} \sqrt{\frac{g}{2h_2}} & 0 \\ 0 & 0 & -\frac{a_3}{A_3} \sqrt{\frac{g}{2h_3}} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \\ x_4 \end{bmatrix} +$$

A X

$$\begin{bmatrix} \frac{\gamma_1 k_1}{A_1} & 0 \\ 0 & \frac{\gamma_2 k_2}{A_2} \\ 0 & \frac{(1-\gamma_2)k_2}{A_3} \\ \frac{(1-\gamma_1)k_1}{A_4} & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

B U

$$\dot{X}' = AX' + BU'$$

(a) Converting to Laplace domain,

$$sX(s) - X(0) = AX(s) + BU(s)$$

$$(sI - A)X(s) = X(0) + BU(s)$$

$$X(s) = (sI - A)^{-1} [X(0) + BU(s)]$$

$$\Phi(s) = (sI - A)^{-1} = \begin{bmatrix} s + \frac{a_1}{A_1} \sqrt{\frac{g}{2h_1}} & 0 & -\frac{a_3}{A_1} \sqrt{\frac{g}{2h_3}} & 0 \\ 0 & s + \frac{a_2}{A_2} \sqrt{\frac{g}{2h_2}} & 0 & -\frac{a_4}{A_2} \sqrt{\frac{g}{2h_4}} \\ 0 & 0 & s + \frac{a_3}{A_3} \sqrt{\frac{g}{2h_3}} & 0 \\ 0 & 0 & 0 & s + \frac{a_4}{A_4} \sqrt{\frac{g}{2h_4}} \end{bmatrix}^{-1}$$

$$h_3^s = \left(\frac{0.4 \times 3.35 \times 3}{0.071 \times \sqrt{18.6}} \right)^2 = (13.128)^2 = 172.344$$

$$h_4^s = \left(\frac{0.3 \times 3.33 \times 3}{0.057 \times \sqrt{18.6}} \right)^2 = (12.191)^2 = 148.620$$

$$h_1^s = \left(\frac{0.7 \times 3.33 \times 3}{\sqrt{18.6 \times 0.071}} + 13.128 \right) = (35.965) = \underline{1293.481}$$

$$h_2^s = \left(\frac{0.6 \times 3.35 \times 3}{\sqrt{18.6 \times 0.057}} + 12.191 \right)^2 = (36.720)^2 = \underline{1348.358}$$

$$a_1 \sqrt{\frac{g}{2h_1^3}} = 4.369 \times 10^{-3}$$

$$a_3 \sqrt{\frac{g}{2h_3^3}} = 0.0119$$

$$a_2 \sqrt{\frac{g}{2h_2^3}} = 3.436 \times 10^{-3}$$

$$a_4 \sqrt{\frac{g}{2h_4^3}} = 0.0103$$

$$(sI - A) = 10^{-3} \begin{bmatrix} s + 0.156 & 0 & 0.425 & 0 \\ 0 & s + 0.107 & 0 & 0.322 \\ 0 & 0 & s + 0.425 & 0 \\ 0 & 0 & 0 & s + 0.322 \end{bmatrix}$$

$$(sI - A)^{-1} = 10^{-3} \times \begin{bmatrix} \frac{1}{s + 0.156} & 0 & \frac{-0.425}{(s + 0.425)(s + 0.156)} & 0 \\ 0 & \frac{1}{s + 0.107} & 0 & \frac{-0.322}{(s + 0.322)(s + 0.107)} \\ 0 & 0 & \frac{1}{s + 0.425} & 0 \\ 0 & 0 & 0 & \frac{1}{s + 0.322} \end{bmatrix}$$

$$\phi(t) = \mathcal{L}^{-1}(\phi(s))$$

$$\phi(t) = \begin{bmatrix} e^{-0.156t} & 0 & \frac{-0.425}{0.269} \times \begin{bmatrix} e^{-0.156t} & -e^{-0.425t} \end{bmatrix} & 0 \\ 0 & e^{-0.107t} & 0 & \frac{-0.322}{0.215} \times \begin{bmatrix} e^{-0.107t} & -e^{-0.322t} \end{bmatrix} \\ 0 & 0 & e^{-0.425t} & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & e^{-0.322t} \end{bmatrix}$$

PROB 2 : Consider dynamic model.

$$\frac{dC_A}{dt} = 10 - C_A - 3.5 \times 10^7 \exp\left(-\frac{6000}{T}\right) C_A$$

$$\frac{dT}{dt} = 298 - 1.3T + 4.2 \times 10^8 \exp\left(-\frac{6000}{T}\right) C_A + 0.3T_c$$

Control obj is to control T by manipulating T_c .

- Plot steady state T_c vs T for $T \in [300, 400]$
- Derive linear state space model for steady state $T = 320$ K
- Compute state transition matrix
- Derive the expression for dynamic response of T for unit step change in T_c

Soln :

(a) For steady state,

$$10 - C_A - 3.5 \times 10^7 \exp\left(-\frac{6000}{T}\right) C_A = 0$$

$$298 - 1.3T + 4.2 \times 10^8 \exp\left(-\frac{6000}{T}\right) C_A + 0.3T_c = 0$$

$$\frac{10}{\left(1 + 3.5 \times 10^7 \exp\left(-\frac{6000}{T}\right)\right)} = C_A$$

$$298 - 1.3T + 4.2 \times 10^8 \exp\left(-\frac{6000}{T}\right) \frac{10}{\left(1 + 3.5 \times 10^7 \exp\left(-\frac{6000}{T}\right)\right)} + 0.3T_c = 0$$

$$0.3T_c = 1.3T - \frac{4.2 \times 10^9 \exp\left(-\frac{6000}{T}\right)}{1 + 3.5 \times 10^7 \exp\left(-\frac{6000}{T}\right)} - 298$$

$$\textcircled{b}) \quad x_1' = C_A - C_A^S \quad u' = T_c - T_c^S$$

$$x_2' = T - T^S$$

$$\frac{dx_1}{dt} = 10 - x_1 - 3.5 \times 10^7 \exp\left(\frac{-6000}{x_2}\right) x_1$$

$$\frac{dx_2}{dt} = 298 - 1.3 x_2 + 4.2 \times 10^8 \exp\left(\frac{-6000}{x_2}\right) x_1 + 0.3 u$$

$$\exp\left(\frac{-6000}{x_2}\right) = \exp\left(\frac{-6000}{x_2^S}\right) + (x_2 - x_2^S) \exp\left(\frac{-6000}{x_2^S}\right) \left(+ \frac{6000}{x_2^{S2}}\right)$$

$$= \exp\left(\frac{-6000}{320}\right) \left[1 + (x_2 - x_2^S) \cdot \left(\frac{6000}{320^2}\right) \right]$$

$$= 7.194 \times 10^{-9} \left[1 + (x_2 - 320) 0.0586 \right]$$

$$\frac{dx_1}{dt} = 10 - x_1 - 0.25179 \left[1 + (x_2 - 320) \cdot 0.0586 \right] x_1$$

$$\frac{dx_2}{dt} = 298 - 1.3 x_2 + 3.02148 \left[1 + (x_2 - 320) \cdot 0.0586 \right] x_1 + 0.3 u$$

PROB 3: Consider the following dynamic model

$$\frac{dc_1}{dt} = \frac{0.5 c_1 c_2}{0.1 + c_2} - u c_1$$

$$\frac{dc_2}{dt} = 4u - u c_2 - \frac{1.25 c_1 c_2}{0.1 + c_2}$$

1. Find optimum value of u to maximize rate of cell production per unit reactor

2. Derive linear state space model using the optimum operating condition.
3. Compute state transition matrix.

SOL:

1. Maximise u_1

At steady state,

$$\frac{0.5 \cancel{C_2}}{0.1 + C_2} = \cancel{u_1} \quad \text{--- (i)}$$

$$4u = uC_2 + \frac{1.25 C_1 C_2}{0.1 + C_2} \quad \text{--- (ii)}$$

$$0.5 C_2 = 0.1u + C_2 u$$

$$C_2 = \frac{0.1u}{0.5 + u}$$

$$4u = \frac{0.1u^2}{0.5 + u} + 1.25 C_1 \left[\frac{\frac{0.1u}{0.5 + u}}{0.1 + \frac{0.1u}{0.5 + u}} \right]$$

$$\cancel{4u} = \frac{0.1 \cancel{u^2}}{0.5 + u} + 1.25 C_1 \left[\frac{\cancel{0.1u}}{0.05 + 0.2u} \right]$$

$$4 = \frac{0.1u}{0.5 + u} + \frac{0.125 C_1}{0.05 + 0.2u}$$

$$4 - \frac{0.1u}{0.5 + u} = \frac{0.125 C_1}{0.05 + 0.2u}$$

$$\left(\frac{0.05 + 0.2u}{0.125} \right) \left[4 - \frac{0.1u}{0.5 + u} \right] = C_1$$

$$u_1 = \left(\frac{0.05u + 0.2u^2}{0.125} \right) \left[4 - \frac{0.1u}{0.5 + u} \right]$$

$$f(u_1) = (0.05 + 0.4u) \left[4 - \frac{0.1u}{0.5 + u} \right] +$$

$$\begin{aligned}
 & \frac{dx}{du} = \dots \\
 & (0.05u + 0.2u^2) \left[\frac{-0.1}{0.5+u} + \frac{0.1u}{(0.5+u)^2} \right] \\
 & > \frac{(0.05+0.4u)(2+3.9u)}{0.5+u} + \frac{u(0.05+0.2u)}{(0.5+u)^2} \left[-0.05 - 0.9u \right] = 0 \\
 & > (0.1 + 0.8u + 0.195u + 1.56u^2)(0.5+u) \\
 & \quad - u \left[0.0025 + 0.07u + 0.045u + 0.18u^2 \right] \\
 & > 0.05 + 0.1u + 0.4u + 0.8u^2 + 0.0975u + 0.195u^2 + 0.78 + 1.56u^3 \\
 & \quad - 0.0025u - 0.055u^2 - 0.18u^3 = 0 \\
 & > 1.38u^3 + 0.94u^2 + 0.595u + 0.83 = 0 \\
 & u^* = -0.9215,
 \end{aligned}$$

$$\frac{dc_1}{dt} = \frac{0.5c_1c_2}{0.1+c_2} - uc_1$$

$$\frac{dc_2}{dt} = 4u - uc_2 - \frac{1.25c_1c_2}{0.1+c_2}$$

$$\begin{aligned}
 x_1 &= c_1, \quad x_2 = c_2 \\
 \frac{dx_1}{dt} &= \left[\frac{0.5x_1^s x_2^s}{0.1+x_2^s} - u^s \right] + \frac{0.5x_2^s}{0.1+x_2^s} (x_1 - x_1^s) - u^s (x_1 - x_1^s) \\
 &+ \left[\frac{0.5x_1}{0.1+x_2^s} - \frac{0.5x_1^s x_2^s}{(0.1+x_2^s)^2} \right] (x_2 - x_2^s) \\
 &- x_1^s (u - u^s)
 \end{aligned}$$

$$\begin{aligned}
 \frac{dx_1'}{dt} &= \left[\frac{0.5x_2^s}{0.1+x_2^s} - u^s \right] (x_1 - x_1^s) + \left[\frac{0.5x_1^s}{0.1+x_2^s} - \frac{0.5x_1^s x_2^s}{(0.1+x_2^s)^2} \right] (x_2 - x_2^s) \\
 &- x_1^s (u - u^s)
 \end{aligned}$$

$$\begin{aligned}
 \frac{dx_2'}{dt} &= -1.25x_2^s (x_1 - x_1^s) + \left[-u^s - \frac{1.25x_1^s}{0.1+x_2^s} + \frac{1.25x_1^s x_2^s}{(0.1+x_2^s)^2} \right] (x_2 - x_2^s)
 \end{aligned}$$

$$\frac{dx_1}{dt} = \frac{0.5x_2^s}{0.1+x_2^s} - u^s + (x_2 - x_2^s)$$

$$+ (4 - x_2^s)(u - u^s)$$

$$\begin{bmatrix} \frac{dx_1'}{dt} \\ \frac{dx_2'}{dt} \end{bmatrix} = \begin{bmatrix} \frac{0.5x_2^s}{0.1+x_2^s} - u^s & \left(\frac{0.5x_2^s}{0.1+x_2^s} \right) \left[1 - \frac{x_2^s}{0.1+x_2^s} \right] \\ -1.25 \frac{x_2^s}{0.1+x_2^s} & -u^s - \frac{1.25x_1^s}{0.1+x_2^s} \left[1 - \frac{x_2^s}{0.1+x_2^s} \right] \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix}$$

$$+ \begin{bmatrix} -x_1^s \\ (4 - x_2^s) \end{bmatrix} u'$$

$$3. \quad \Phi(s) = (sI - A)^{-1}$$

PROB 4: Consider dynamic model of reactor

$$\frac{dC_A}{dt} = 10u - \left(u + \frac{5}{6}\right)C_A - \frac{1}{6}C_A^2$$

$$\frac{dC_B}{dt} = \frac{5}{6}C_A - \left(\frac{5}{3} + u\right)C_B$$

1. Find optimum u to maximize production of B.
2. Derive linear state space model using steady state value of $u=2$
3. Compute state transition matrix.

SOL:

1. At steady state, $\frac{dC_A}{dt} = 0$; $\frac{dC_B}{dt} = 0$

$$10u - \left(u + \frac{5}{6}\right)C_A - \frac{C_A^2}{6} = 0 \quad \text{---(i)}$$

$$\frac{5}{6}C_A - \left(\frac{5}{3} + u\right)C_B = 0 \quad \text{---(ii)} \quad -11 \pm \sqrt{121 + 240u}$$

Eqn (i),

$$C_A^2 + [6u + 5]C_A - 60u = 0$$

$$C_A = \frac{- (6u + 5) \pm \sqrt{(6u + 5)^2 + 4 \cdot 60u}}{2} \quad [C_A > 0]$$

Substituting C_A in eqn (ii),

$$\frac{5}{6} \left[\frac{- (6u + 5) + \sqrt{(6u + 5)^2 + 240u}}{2} \right] = \left(\frac{5}{3} + u\right) C_B$$

$$\Rightarrow C_B = \frac{\frac{5}{12} \left[- (6u + 5) + \sqrt{(6u + 5)^2 + 240u} \right]}{\left(\frac{5}{3} + u\right)}$$

$\frac{dC_B}{du} = 0$ for maximum C_B .

$$\left[-6 + \frac{2(6u + 5) \cdot 6 + 240}{2\sqrt{(6u + 5)^2 + 240u}} \right] - \frac{45}{\left(\frac{5}{3} + u\right)^2} = 0$$

(5/3+u)

$$(5/3+u) \left[-6 + \frac{12(6u+5) + 240}{2N} \right] - N = 0$$

$$(5/3+u) \left[-12N + \frac{12(6u+5) + 240}{1} \right] - 2N^2 = 0$$

$$\frac{5}{3} [-12N + \{72u + 300\}] - 2[(6u+5)^2 + 240u] = 0$$

$$+ u [-12 + \{72u + 300\}]$$

$$-20N + \{140u + 500\} - 2[36u^2 + 300u + 25]$$

$$-12u + \{72u^2 + 300u\} = 0$$

$$-20N - 172u + 450 = 0$$

$$10N = 86u - 225$$

$$100 \left[\sqrt{(6u+5)^2 + 240u} \right]^2 = (86u - 225)^2$$

$$(6u+5)^2 + 240u = 0.7396u^2 + 506.25 - 387u$$

$$36u^2 + 25 + 300u = 0.7396u^2 + 506.25 - 387u$$

$$35.2604u^2 - 481.25 + 687u = 0$$

$$u_{opt} = 0.676, -20.160(x)$$

$$2. \quad \frac{dC_A}{dt} = 10u - \left(u + \frac{5}{6}\right)C_A - \frac{C_A^2}{6}$$

$$\frac{dC_B}{dt} = \frac{5C_A}{6} - \left(\frac{5}{3} + u\right)C_B$$

$$x_1 = C_A \quad x_2 = C_B \quad u = 2$$

$$x_1' = C_A - C_A' \quad x_2' = C_B - C_B'$$

$$C_A^s = \frac{-(6u+5) + \sqrt{(6u+5)^2 + 240u}}{2}$$

$$= 5.365$$

$$C_B^S = \frac{5/6 C_A}{(5/3 + u)} = 1.219$$

$$\frac{dx_1}{dt} = -\frac{x_1^2}{6} - \frac{5x_1}{6} - x_1 \cdot u + 10u$$

$$\frac{dx_2}{dt} = \frac{5}{6}x_1 - \frac{5}{3}x_2 - u \cdot x_2$$

$$\frac{dx_1^S}{dt} = 0 = -\frac{x_1^S^2}{6} - \frac{5x_1^S}{6} - x_1^S u^S + 10u^S$$

$$\frac{dx_2^S}{dt} = 0 = \frac{5}{6}x_1^S - \frac{5}{3}x_2^S - u^S \cdot x_2^S$$

$$\begin{aligned} \frac{dx_1}{dt} = & \left[-\frac{x_1^S^2}{6} - \frac{5x_1^S}{6} - x_1^S \cdot u^S + 10u^S \right] + \\ & (x_1 - x_1^S) \left[-\frac{2x_1^S}{6} - \frac{5}{6} - u^S \right] + \\ & (u - u^S) \left[-x_1^S + 10 \right] \end{aligned}$$

$$\frac{dx_1'}{dt} = x_1' \left[-\frac{2x_1^S}{6} - \frac{5}{6} - u^S \right] + u_1' \left[-x_1^S + 10 \right]$$

$$\begin{aligned} \frac{dx_2}{dt} = & \left[\frac{5x_1^S}{6} - \frac{5x_2^S}{3} - u^S x_2^S \right] + x_1' \left[\frac{5}{6} \right] \\ & + x_2' \left[-5/3 - u^S \right] + u' \left[x_2^S \right] \end{aligned}$$

$$\frac{dx_2'}{dt} = \frac{5}{6}x_1' + \left[-5/3 - u^S \right] x_2' + x_2^S u'$$

$$u' = \left[\frac{5}{6}x_1^S - \frac{5}{3}x_2^S - u^S \right] \cdot \left[x_1' \right] \cdot \left[-x_1^S + 10 \right]$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} -22 & 5/6 \\ 5/6 & -5/3 - u^* \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} + \begin{bmatrix} x_2^* \\ x_2^* \end{bmatrix} u'$$

$$\dot{x} = Ax' + Bu'$$

Substituting values,

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} -13.563 & 0 \\ 0.833 & -3.667 \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} + \begin{bmatrix} 4.635 \\ 1.219 \end{bmatrix} u'$$

$$\Phi(s) = (sI - A)^{-1} = \begin{bmatrix} s + 13.563 & 0 \\ 0.833 & s + 3.667 \end{bmatrix}^{-1}$$

$$= \frac{1}{(s + 13.563)(s + 3.667)} \begin{bmatrix} s + 3.667 & 0 \\ -0.833 & s + 13.563 \end{bmatrix}$$

$$\Phi(t) = \mathcal{L}^{-1}\{\Phi(s)\}$$

$$= \mathcal{L}^{-1} \left\{ \begin{bmatrix} \frac{1}{s + 13.563} & 0 \\ \frac{-0.833}{(s + 13.563)(s + 3.667)} & \frac{1}{s + 3.667} \end{bmatrix} \right\}$$

$$= \begin{bmatrix} e^{-13.563t} & 0 \\ 0.817 \begin{bmatrix} e^{-13.563t} & -e^{-3.667t} \end{bmatrix} e^{-3.667t} \end{bmatrix}$$

PROB 5: Transfer function:

$$G(s) = \frac{180}{(s+2)(s+3)(s+4)(s+5)}$$

- (a) Find an equivalent first order dead time (FODT) model using moment method
- (b) Find an " second order dead time (SODT) having equal time constants $\frac{Ke^{-\theta s}}{(\tau s+1)^2}$ using moment method.

SOL:

$$G(s) = \frac{180}{(s+2)(s+3)(s+4)(s+5)}$$

$$G(0) = \frac{180}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{3}{2} = 1.5$$

$$H(s) = \frac{Ke^{-\theta s}}{\tau s+1} \quad H(0) = K = 3/2 = 1.5$$

$$G'(s) = \frac{-180}{(s+2)(s+3)(s+4)(s+5)} \left[\frac{1}{s+2} + \frac{1}{s+3} + \frac{1}{s+4} + \frac{1}{s+5} \right]$$

$$G'(s) = -G(s) \left[\frac{1}{s+2} + \frac{1}{s+3} + \frac{1}{s+4} + \frac{1}{s+5} \right]$$

$$H'(s) = \frac{-\theta Ke^{-\theta s}}{\tau s+1} - \frac{\tau Ke^{-\theta s}}{(\tau s+1)^2} = -\frac{Ke^{-\theta s}}{\tau s+1} \left[\theta + \frac{\tau}{\tau s+1} \right]$$

$$= -H(s) \left[\theta + \frac{\tau}{\tau s+1} \right]$$

$$G''(s) = +G(s) \left[\frac{1}{(s+2)^2} + \frac{1}{(s+3)^2} + \frac{1}{(s+4)^2} + \frac{1}{(s+5)^2} \right] - G'(s) \left[1 \dots \right]$$

$$H''(s) = +H(s) \left[\frac{\tau^2}{(\tau s+1)^2} \right] - H'(s) \left[\theta + \frac{\tau}{\tau s+1} \right]$$

$$G'(0) = -1.5 \left[\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right] = -1.5 [1.2834]$$

$$[\dots] = -1.5(\theta + \tau)$$

$$H'(0) = -K[\theta + \tau] = -1.5[0.6024 + 0.681] = -1.5(1.2834)$$

$$\theta + \tau = 1.2834 \quad \text{--- (i)}$$

$$G''(0) = 1.5 \left[\frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} \right] + 1.5 [1.2834] \left[\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right]$$

$$G''(0) = 1.5 \left[0.4636 + 1.2834^2 \right]$$

$$H''(0) = K\tau^2 + K(\tau + \theta)(\theta + \tau)$$

$$= K[\tau^2 + (\tau + \theta)^2]$$

$$1.5 [0.4636 + 1.2834^2] = 1.5 [\tau^2 + (\tau + \theta)^2] \quad \text{--- (ii)}$$

$$\tau^2 = 0.4636 \Rightarrow \tau = 0.681$$

$$\tau + \theta = 1.2834 \Rightarrow \theta = 0.6024$$

$$\boxed{K = 1.5 ; \tau = 0.681 ; \theta = 0.6024}$$

PROB 7: Transfer function = $G(s) = \frac{120(s+6)(s+7)}{(s+2)(s+3)(s+4)(s+5)}$

Derive state space model in

- (a) controllable canonical form
- (b) Jordan canonical form
- (c) Observable canonical form